ESP32 Drone Detection System

Executive Overview

What it does: A credit-card-sized microcontroller detects drones passively by listening to their Wi-Fi signals. No radar. No jamming. Just listening.

The insight: Consumer drones broadcast constant Wi-Fi management frames (beacon signals). An ESP32 in promiscuous mode can hear these frames from any drone within Wi-Fi range — and determine with 80–100% confidence whether the device is a drone or just someone's phone.

Metric	Value	Context
Hardware Cost	$5–$15	ESP32 dev board + USB cable
Power Draw	~0.2W	Runs on phone battery for days
Detection Range	50–150m	Line-of-sight Wi-Fi range
Confidence	0–100%	4-factor scoring model
Memory Footprint	~5 KB	50 devices tracked simultaneously
Legal Risk	Zero	Passive receive-only. No transmission.

How It Works (3-Step Process)

Step 1 — Listen

The ESP32 places its Wi-Fi radio into promiscuous mode — the same mode network analysts use for packet sniffing. It cycles through all 13 Wi-Fi channels (2.4 GHz) every 2.6 seconds, capturing every frame in range regardless of its destination. No handshake, no login, no transmission.

Step 2 — Identify

For each detected device, the system extracts four signals:

Signal	What It Measures	Drone Tell
Vendor OUI	MAC address prefix	DJI/Parrot/Autel have known prefixes
Packet Rate	Frames per second	Drones stream video >30 pkts/sec
RSSI Stability	Signal strength variance	Hovering drone = stable RSSI
SSID	Network name broadcast	SSIDs like "DJI-Mavic-Pro"

Step 3 — Score

These four signals feed a weighted scoring model that produces a 0–100% confidence score every second. **Vendor OUI (40 pts)** is the heaviest signal because MAC prefixes are hardware-burned and cannot be spoofed in real time. **Packet rate (30 pts)** catches active video links. **RSSI stability (20 pts)** separates a hovering drone from a walking person. **SSID keywords (10 pts)** catches drones broadcasting their model name.

What the Output Looks Like

The system outputs one report per second over a USB serial connection. Each line lists every device it sees, its vendor, signal strength, and confidence score. **No app, no dashboard, no cloud connection needed.**

```
--- ESP32 Drone Detector ---
SSID: DJI-Mini-3Pro | Vendor: DJI | 42 pkt/s | RSSI: -61 dBm | Var: 2.14 | CONF: 90%
| MAC: A4:12:6D:xx
SSID: (hidden) | Vendor: TP-Link | 5 pkt/s | RSSI: -72 dBm | Var: 8.31 | CONF: 0% |
MAC: 88:3A:DD:xx
No drone signatures detected.
Frames: 1,247 | Channel: 7/13 | Slots: 2/50
------------------------------
```

The confidence scale:

Score	Meaning	Action
80–100%	Almost certainly a drone	Alert
50–79%	Likely a drone	Flag for review
10–49%	Possible drone	Track only
0–9%	Ambient Wi-Fi device	Ignore

Deployment — What You Need

Hardware: One ESP32 dev board (any variant with WiFi — ~$5–$15 on Amazon or Alibaba) and a USB power source. Optional: an external 2.4 GHz antenna for extended range.

Setup time: 15 minutes. Plug in, flash the firmware via Arduino IDE, open serial monitor. That is the entire deployment process.

Form factor: The ESP32 is the size of a credit card and runs on a phone charger. It can be deployed in a weather box, taped to a window, or mounted on a pole with a USB battery pack.

Cost Comparison vs. Commercial C-UAS

Solution	Per-Node Cost	Power	Deployment
Commercial C-UAS	$2,000–$50,000+	50–500W	Permanent install, FCC licensing
ESP32 Passive Sniffer	$5–$15	0.2W	Battery-powered, zero paperwork
Multi-node Network (10x)	$50–$150	2W	Cover a facility perimeter

Caveat: The ESP32 solution does not replace high-end C-UAS. It does not jam, intercept, or geolocate. What it does is **detect with high confidence at essentially zero marginal cost** — enabling perimeter awareness where no detection capability currently exists.

Roadmap

Phase	Capability	Timeline
1 — Current	Single-node, 2.4 GHz, 4 drone vendors	Now
2 — OUI Expansion	50+ drone vendor fingerprints, 5 GHz support	Week 1–2
3 — Multi-Node	3+ ESP32s triangulating drone position	Month 2
4 — ML Classification	TinyML model on RSSI time-series	Month 3
5 — Protocol Detection	MAVLink/CRSF/ELRS signature recognition	Month 4

The Bottom Line

This is a **$5 sensor** that answers a $50,000 question: *"Is there a drone nearby?"* It is not a full counter-UAS system. It is a **force multiplier** — a passive, always-on, zero-footprint early warning sensor that scales to cover perimeters, critical infrastructure, and event venues at commodity hardware prices.

Source: Reverse-engineered from Patel et al. — "ESP32-Based Passive Wi-Fi Drone Detection with Confidence-Weighted Fingerprinting," IJRPR Vol 7(4), Apr 2026. Implementation and modifications by C4C Engineering.